

UNIT III: PROTOCOLS AND NETWORK MODELS

Topics Covered

- **3.1** Network Protocols: TCP/IP, SMTP
- **3.2** Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS)
- **3.3** OSI Reference Model (7 Layers)
- **3.4** TCP/IP Reference Model (4/5 Layers)
- **3.5** Switching Techniques: Circuit Switching, Message Switching, and Packet Switching

3.1 NETWORK PROTOCOLS: TCP/IP & SMTP

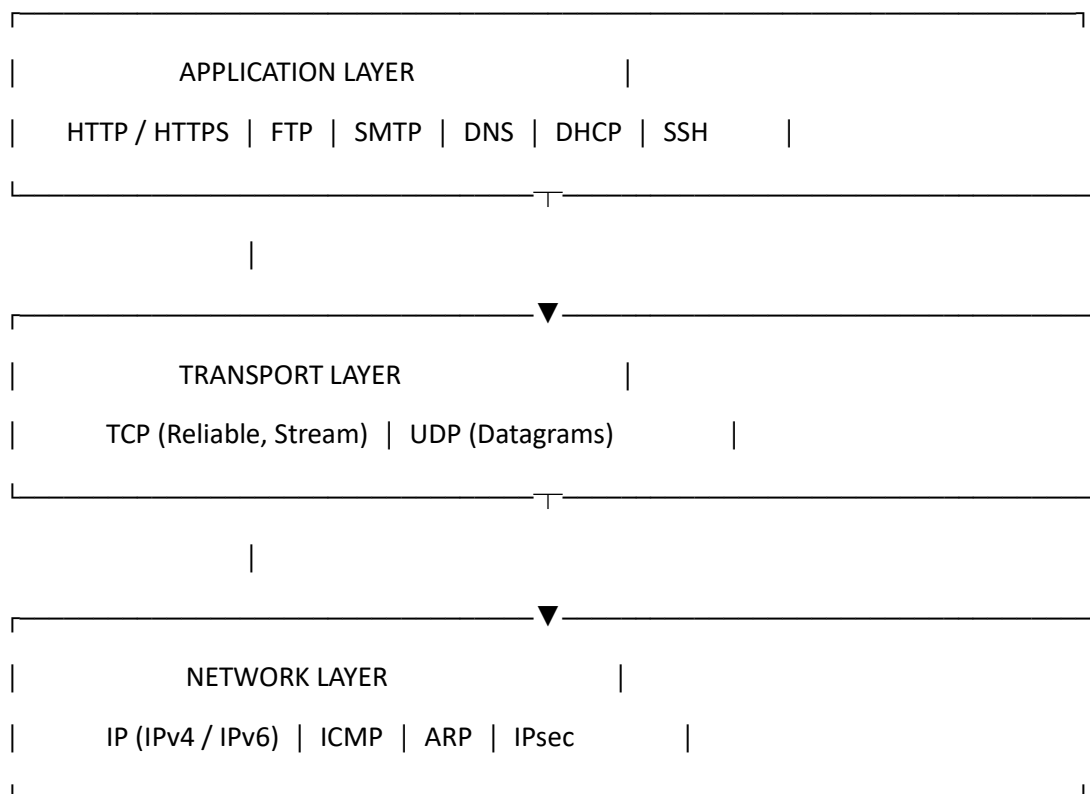
What is a Protocol?

A **network protocol** is a formal set of rules, standards, and conventions that govern how data is formatted, transmitted, routed, received, and processed across connected devices on a network.

Without protocols, devices would be unable to interpret the structure or sequence of transmitted signals.

1. TCP/IP Protocol Suite Overview

The **TCP/IP Suite** consists of foundational protocols operating at different layers of the network architecture.



A. Transmission Control Protocol (TCP)

TCP is a core **Transport Layer** protocol designed for reliable, connection-oriented byte-stream delivery.

- **Port Numbers:** Default ports include 80 (HTTP), 443 (HTTPS), 21 (FTP), 25 (SMTP).
- **Key Characteristics:**
 - **Reliability via Acknowledgments:** Receivers send ACK packets back to the sender. If an ACK is not received within a timeout window, the segment is retransmitted.
 - **Sequence Control:** Assigns sequence numbers to every byte transmitted, allowing the receiver to reassemble out-of-order packets.
 - **Flow Control:** Utilizes a **Sliding Window Protocol** to ensure a sender does not overwhelm a slow receiver.
 - **Congestion Control:** Employs algorithms (e.g., Slow Start, Congestion Avoidance) to prevent network collapse.

B. Internet Protocol (IP)

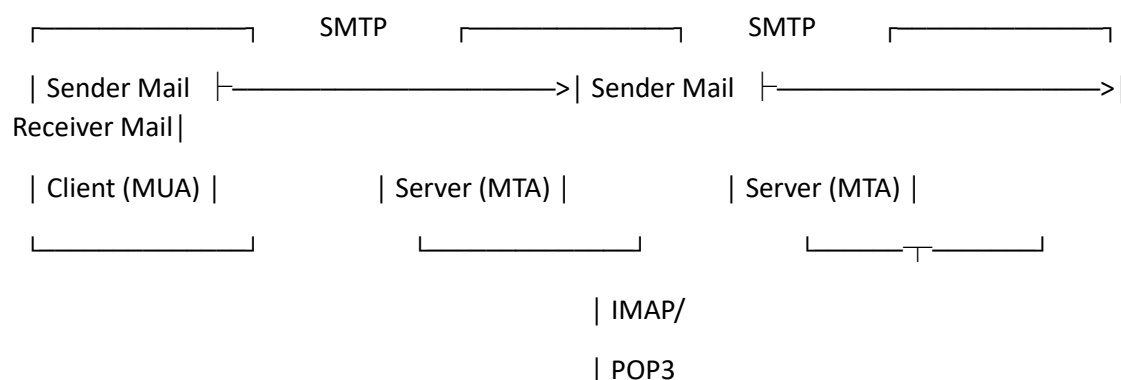
IP is the primary **Network Layer** protocol responsible for logical addressing and packet routing across network boundaries.

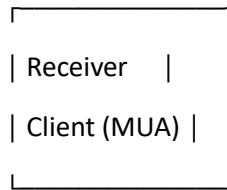
- **Key Characteristics:**
 - **Connectionless & Best-Effort:** Does not establish a prior connection and does not guarantee delivery, packet order, or error correction (delegated to upper layers).
 - **Packet Encapsulation:** Encapsulates Transport layer segments into IP datagrams containing source and destination IP addresses.
 - **Fragmentation & Reassembly:** Splits packets if they exceed the Maximum Transmission Unit (MTU) of an intermediate link.

2. Simple Mail Transfer Protocol (SMTP)

SMTP is an **Application Layer** protocol used to send and relay outgoing electronic mail across IP networks.

- **Standard Port:** Port 25 (unencrypted relaying), Port 587 (secure submission via STARTTLS), Port 465 (legacy SMTPS).
- **Protocol Type:** Text-based, connection-oriented, and operates on top of **TCP**.





How Mail Delivery Works

1. **User Agent (MUA):** The sender drafts an email and uses SMTP to submit it to their local Mail Transfer Agent (MTA) server.
2. **Mail Transfer Agent (MTA):** The sender's mail server looks up the recipient's domain using DNS **MX (Mail Exchange)** records.
3. **Server-to-Server Relay:** The sender's MTA establishes a TCP connection with the recipient's MTA on Port 25 and transfers the message using SMTP commands.
4. **Final Retrieval:** The recipient retrieves the message from their mail server using **POP3** or **IMAP** (SMTP is only used for *sending/relaying*, not for client retrieval).

Basic SMTP Command-Response Sequence

HTTP

Client: HELO mail.sender.com

Server: 250 Hello mail.sender.com

Client: MAIL FROM:<alice@sender.com>

Server: 250 OK

Client: RCPT TO:<bob@receiver.com>

Server: 250 OK

Client: DATA

Server: 354 Start mail input; end with <CR><LF>.<CR><LF>

Client: Subject: Meeting Notice

Client: Hi Bob, see you at 10 AM.

Client: .

Server: 250 Message accepted for delivery

Client: QUIT

Server: 221 Service closing transmission channel

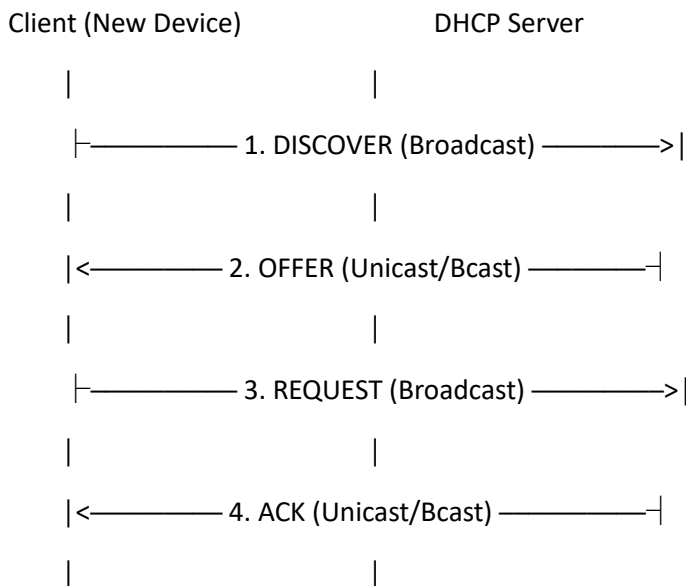
3.2 DHCP AND DNS

1. Dynamic Host Configuration Protocol (DHCP)

DHCP is an **Application Layer** protocol that automates the network configuration of devices connected to an IP network.

- **Transport Protocol:** Operates over **UDP** using Port 67 (Server) and Port 68 (Client).
- **Function:** Dynamically assigns IP addresses, Subnet Masks, Default Gateways, DNS Server addresses, and Lease Durations to clients.

The DORA Process (4-Way Handshake)



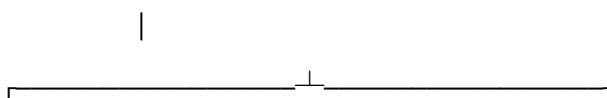
1. **Discover:** The newly connected client broadcasts a DHCPDISCOVER frame (255.255.255.255) onto the local network segment asking for an IP address.
2. **Offer:** DHCP server(s) receive the request and reply with a DHCPOFFER packet containing a reserved IP address, subnet mask, gateway, and lease duration.
3. **Request:** The client broadcasts a DHCPREQUEST message acknowledging acceptance of the offer from a specific DHCP server.
4. **Acknowledge:** The chosen DHCP server sends a DHCPACK packet confirming the IP lease assignment. The client configures its interface.

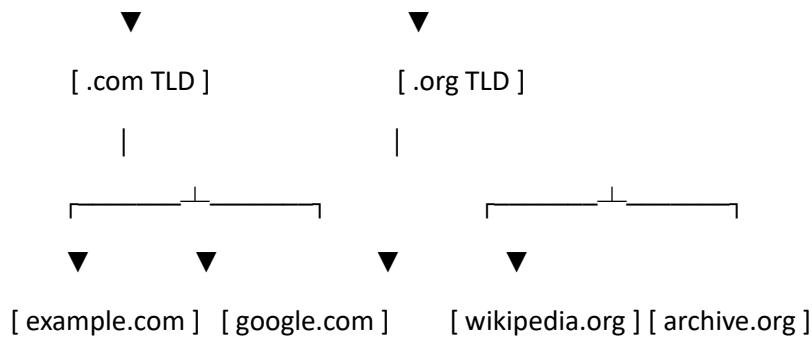
2. Domain Name System (DNS)

DNS is an **Application Layer** administrative database system that translates human-readable domain names (e.g., www.example.com) into machine-routable numerical IP addresses (e.g., 93.184.216.34).

- **Transport Protocol:** Uses **UDP Port 53** for standard client queries (fast, low overhead) and **TCP Port 53** for zone transfers between DNS servers or long payloads.
- **Architecture:** Formed as a hierarchical, distributed database tree.

[Root Server "."]





The DNS Hierarchy Layers

1. **Root Name Servers (.):** Top of the hierarchy (13 root IP address clusters worldwide). Directs queries to the correct TLD server.
2. **Top-Level Domain (TLD) Servers:** Manages domain extensions such as .com, .org, .net, .edu, and country codes (.in, .uk).
3. **Authoritative Name Servers:** Holds the official, final resource records (A, AAAA, MX, CNAME) for specific domain names configured by domain registrars.
4. **Recursive Resolvers:** Usually operated by ISPs or public DNS providers (e.g., 8.8.8.8, 1.1.1.1). Performs recursive searches on behalf of local endpoints.

Common DNS Resource Records

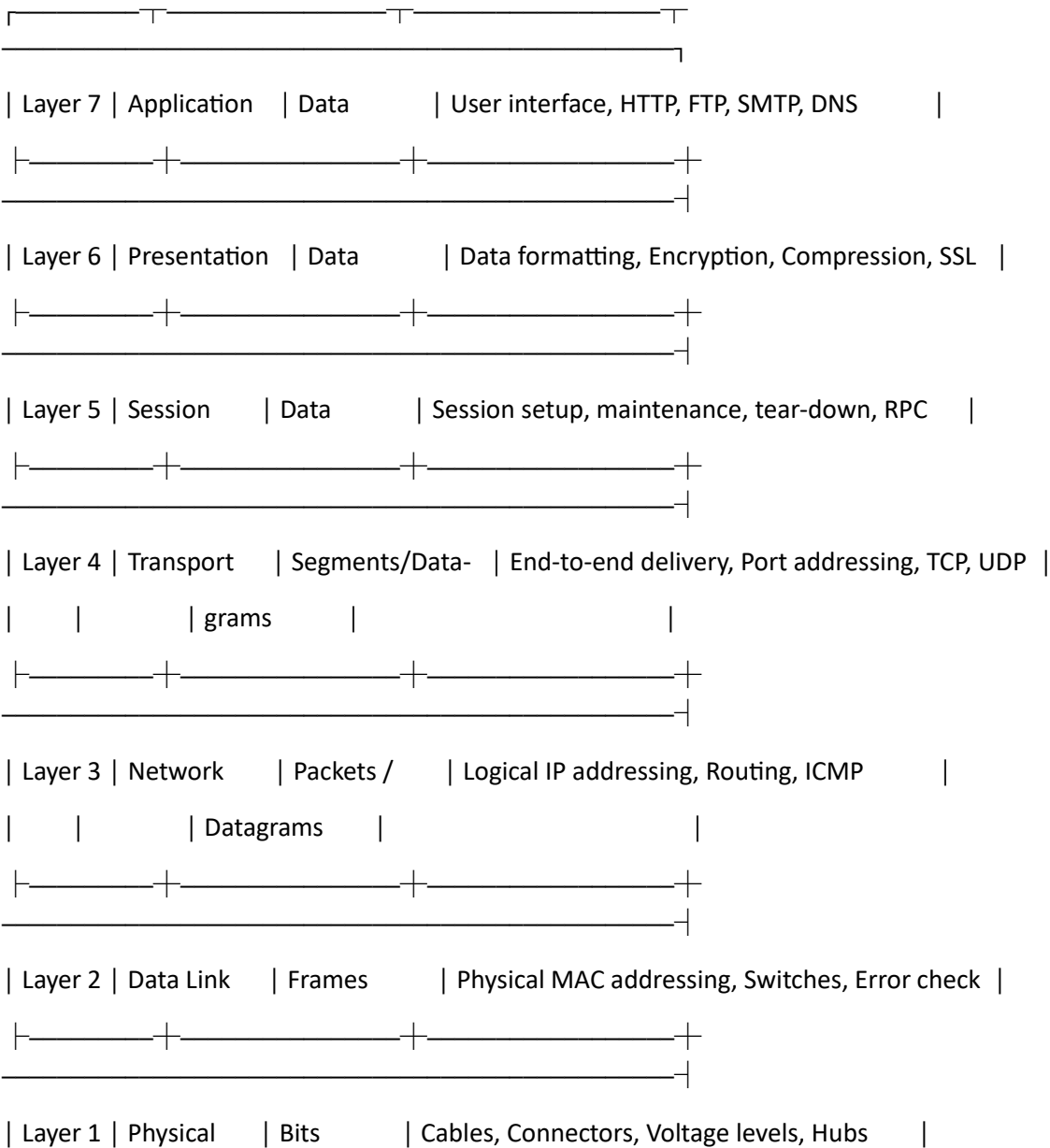
Record Type	Full Name	Purpose / Function
A	Address Record	Maps a domain name to an IPv4 address
AAAA	IPv6 Address Record	Maps a domain name to an IPv6 address
CNAME	Canonical Name	Creates an alias pointing one domain name to another domain name
MX	Mail Exchange	Points to the primary mail servers responsible for accepting email for the domain
NS	Name Server	Specifies the authoritative DNS servers for a domain
TXT	Text Record	Holds arbitrary text; used for domain security checks (SPF, DKIM, DMARC)

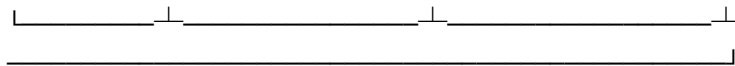
Record Type	Full Name	Purpose / Function
PTR	Pointer Record	Used for reverse DNS lookups (maps an IP address back to a hostname)

3.3 OSI REFERENCE MODEL (ISO/IEC 7498-1)

Developed by the International Organization for Standardization (ISO), the **Open Systems Interconnection (OSI)** model is a conceptual 7-layer framework detailing standard network communications.

Layer # Layer Name Data Unit (PDU) Primary Functions / Protocols





Layer-by-Layer Breakdown

Layer 7: Application Layer

- **Role:** Provides network interfaces directly to end-user software applications.
- **Functions:** Resource identification, network service access.
- **Protocols/Devices:** HTTP, HTTPS, FTP, SMTP, SSH, Telnet, DNS.

Layer 6: Presentation Layer

- **Role:** Handles syntax, formatting, and translation of transmitted data so upper systems understand it.
- **Functions:** Data translation (ASCII to EBCDIC), Data Compression, Data Encryption and Decryption (SSL/TLS).
- **Formats:** JPEG, PNG, MP3, ASCII, MIME.

Layer 5: Session Layer

- **Role:** Establishes, manages, synchronizes, and terminates interactive sessions between applications.
- **Functions:** Dialog control (Simplex, Half-Duplex, Full-Duplex), session checkpointing and recovery.
- **Protocols:** NetBIOS, RPC (Remote Procedure Call), PPTP.

Layer 4: Transport Layer

- **Role:** Ensures end-to-end transmission of data streams between processes across nodes.
- **Functions:** Service-point (Port) addressing, segmentation & reassembly, flow control, connection management, error control.
- **Protocols:** TCP, UDP, SCTP.

Layer 3: Network Layer

- **Role:** Routes data packets across multiple interconnected networks from source host to destination host.
- **Functions:** Logical Addressing (IPv4/IPv6), Path Determination (Routing), Traffic Control.
- **Protocols/Devices:** IP, ICMP, ARP, OSPF, BGP; **Routers**, Layer 3 Switches.

Layer 2: Data Link Layer

- **Role:** Provides reliable node-to-node frame transfer over a physical medium.
- **Sublayers:**
 1. **Logical Link Control (LLC):** Manages flow/error control and interfaces with upper layer protocols (IEEE 802.2).

2. **Media Access Control (MAC):** Appends physical MAC addresses and manages medium access (IEEE 802.3/802.11).

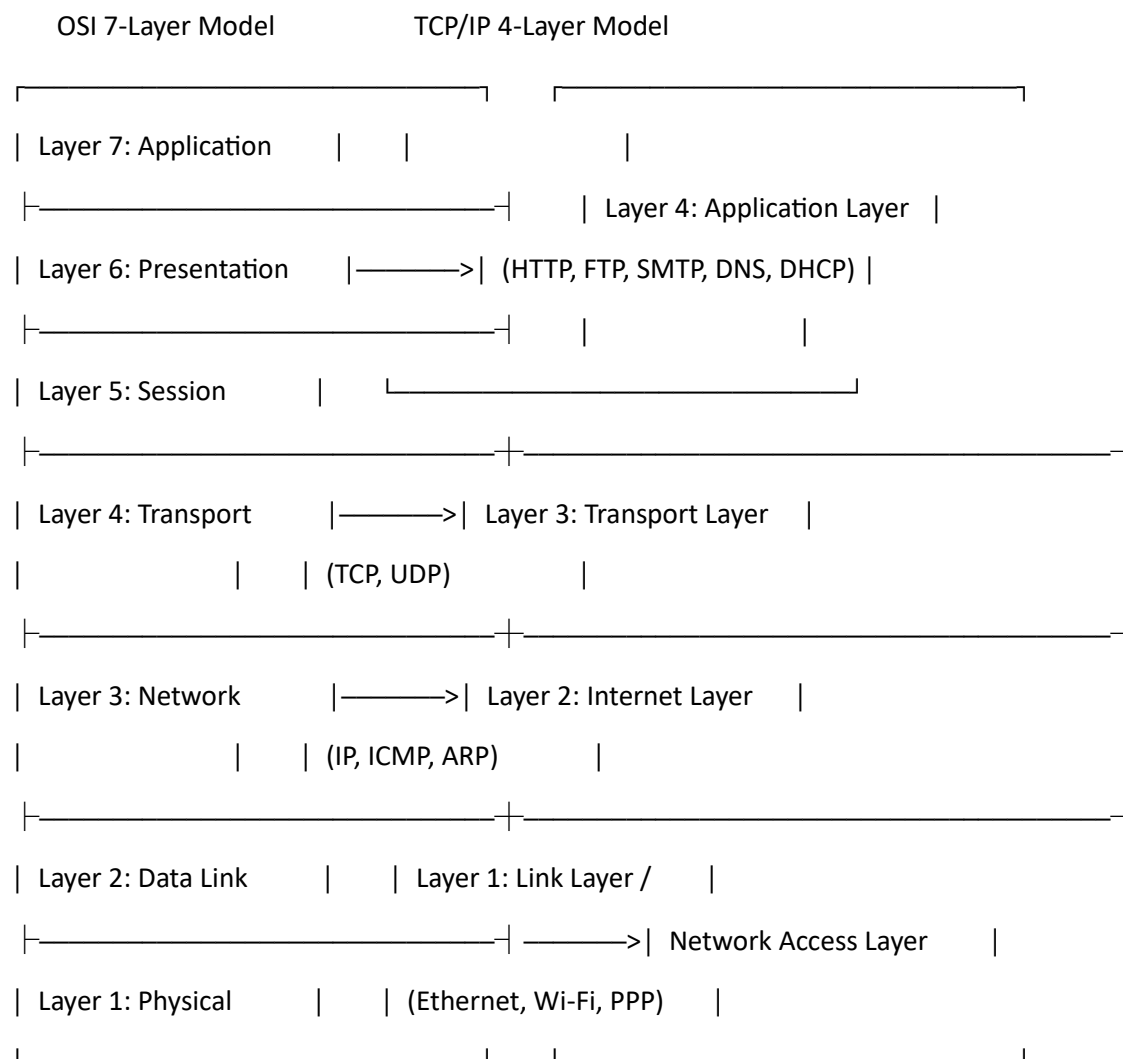
- **Functions:** Physical addressing (MAC), Framing, Error Detection (CRC/FCS), Flow Control.
- **Devices:** **Network Switches**, Bridges, Network Interface Cards (NICs).

Layer 1: Physical Layer

- **Role:** Transmits unstructured raw bit streams (0s and 1s) over physical communication channels.
- **Functions:** Pin layout definitions, bit rate control, voltage thresholds, line coding, cable specifications.
- **Devices:** Ethernet Cables (UTP/STP), Fiber Optics, **Hubs**, Repeaters, Connectors (RJ-45).

3.4 TCP/IP REFERENCE MODEL

The **TCP/IP Model** (also known as the Internet Architecture Model) was created by the US Department of Defense (DoD). It is a practical 4-layer model (or updated 5-layer version) built around real-world deployment.

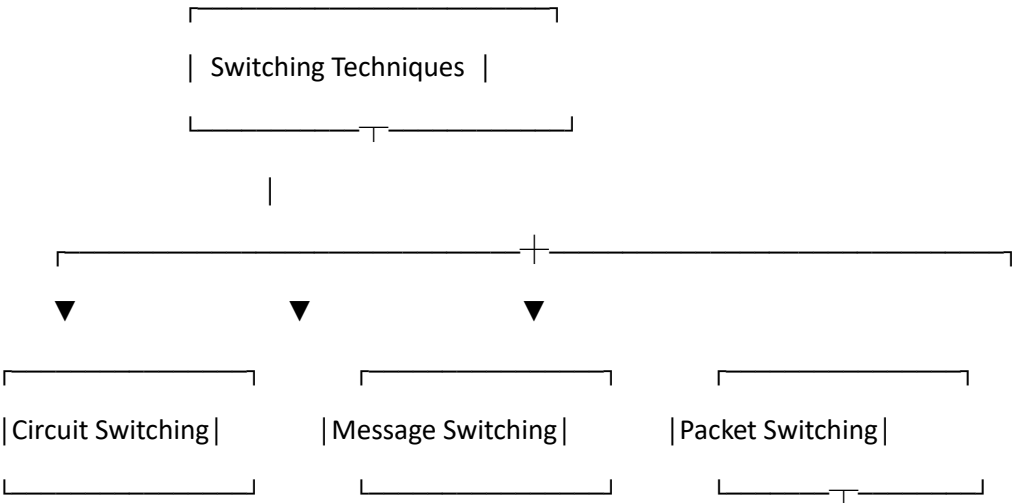


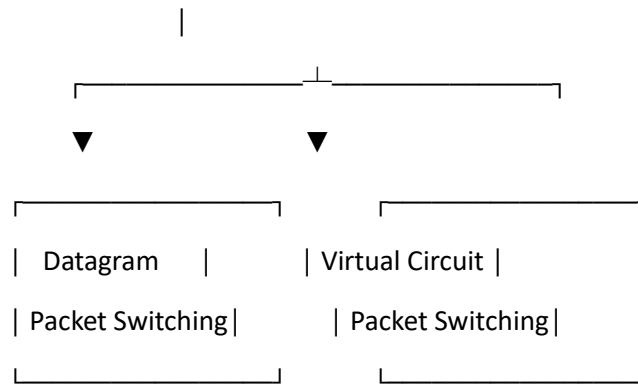
Comprehensive Comparison: OSI vs. TCP/IP Model

Comparison Factor	OSI Reference Model	TCP/IP Reference Model
Origin & Purpose	Developed by ISO as a theoretical reference guide	Developed by DoD / DARPA for real-world Internet implementation
Number of Layers	7 Layers	4 Layers (or 5 in updated academic literature)
Session & Presentation	Distinct layers (Layers 5 & 6)	Combined into the single Application Layer
Protocol Approach	Model was defined <i>before</i> protocols were invented	Protocols were created <i>first</i> ; model is a description of protocols
Transport Layer Support	Supports both Connection-Oriented & Connectionless	Supports both Connection-Oriented (TCP) & Connectionless (UDP)
Network Layer Support	Supports both Connection-Oriented & Connectionless	Supports only Connectionless (IP)
Industry Adoption	Used primarily as a reference and teaching tool	The actual dominant architecture powering the Internet

3.5 SWITCHING TECHNIQUES

Switching is the mechanism by which nodes connect and route data through intermediate nodes across a network topology.





1. Circuit Switching

Establishes a **dedicated physical communication path** between sender and receiver before data transmission begins.

- **Phases:**
 1. Circuit Setup
 2. Data Transfer
 3. Circuit Teardown
- **Characteristics:** Guaranteed bandwidth, fixed channel capacity, zero middle-node delay during transfer.
- **Disadvantages:** Low resource utilization (channel remains reserved even during silences/inactivity).
- **Examples:** Traditional Landline PSTN telephone networks.

2. Message Switching

Transmits data as complete, variable-length blocks called **messages**.

- **Store-and-Forward:** Intermediate nodes receive an entire message, store it locally on disc/memory, inspect headers, and forward the complete message to the next hop.
- **Characteristics:** No setup phase required; dynamic channel utilization.
- **Disadvantages:** Extremely high delay, requires massive buffer storage on intermediate switches; unsuitable for real-time traffic.
- **Examples:** Historical Telegraph systems, early electronic military message routing.

3. Packet Switching

Breaks large messages into small, fixed or bounded variable-sized chunks called **packets**. Each packet carries source/destination addresses and sequence numbers. Packets travel independently through the network.

Variations of Packet Switching

1. **Datagram Packet Switching (Connectionless):**

- Each packet is treated independently.
- Intermediate routers evaluate destination addresses and select the best path for each packet dynamically.
- Packets may arrive out of order and must be reassembled at the destination.
- *Example:* Standard Internet Protocol (IP) routing.

2. Virtual Circuit Packet Switching (Connection-Oriented):

- A logical path (Virtual Circuit) is selected through the network before sending packets.
- All packets follow the pre-established route sequentially.
- *Example:* Frame Relay, ATM (Asynchronous Transfer Mode), MPLS networks.

Detailed Comparison Matrix of Switching Technologies

Feature Parameter	Circuit Switching	Message Switching	Packet Switching
Path Establishment	Mandatory setup phase	No setup phase	No setup (Datagram) / Logical setup (Virtual Circuit)
Data Unit Format	Continuous signal / bitstream	Complete, variable message block	Small, bounded packets
Intermediate Storage	No node storage (real-time stream)	Store-and-Forward (entire message)	Store-and-Forward (individual packets)
Bandwidth Reservation	Dedicated / Reserved	Shared dynamically	Shared dynamically
Transmission Delay	Uniform low delay post-setup	High, unpredictable delay	Low-to-moderate delay
Buffer Memory Demand	Minimal	High memory/disk requirements	Moderate router memory (RAM)
Resource Efficiency	Low (idle time wastes line)	High utilization	Highest utilization

Feature Parameter	Circuit Switching	Message Switching	Packet Switching
Primary Real-World Use	PSTN landline calls	Telegraphy, historical messaging	Internet data traffic, modern networks

QUICK REVISION & SUMMARY

Essential Technical Terms

- **TCP:** Connection-oriented transport protocol offering reliable byte-stream transfer via acknowledgments and windowing.
- **SMTP:** Application layer text protocol for sending/relaying email using TCP Port 25.
- **DHCP:** Protocol that automates client IP, gateway, and DNS configuration via the **DORA** handshake.
- **DNS:** Hierarchical database system resolving domain names to IP addresses via resource records (A, AAAA, MX, CNAME).
- **OSI Model:** A theoretical 7-layer framework defining network tasks from Physical bit transmission up to the Application interface.
- **TCP/IP Model:** A practical 4-layer architecture powering the Internet.
- **Packet Switching:** Breaking data into independent units called packets to maximize line utilization across shared paths.

IMPORTANT EXAM QUESTIONS

Short Answer Questions

1. Define the 4 steps of the DHCP **DORA** operational process.
2. What are the primary functions of DNS **A**, **AAAA**, **MX**, and **CNAME** records?
3. State the primary differences between Layer 2 (Data Link) and Layer 3 (Network) addressing.
4. What is the role of the Presentation Layer in the 7-layer OSI model?
5. Why is Packet Switching more bandwidth-efficient than Circuit Switching?
6. Which Transport layer protocol does DNS use, and under what conditions does it switch between them?

Long Answer Questions

1. **Application Layer Protocols:** Explain the architectural operation of SMTP mail delivery. Show how an email moves from client user-agent through MTAs to the recipient, including basic SMTP commands.
2. **OSI Model Deep-Dive:** Detail all 7 layers of the OSI reference model. State the Protocol Data Unit (PDU), primary responsibilities, protocols, and hardware associated with each layer.

3. **Model Comparison:** Provide a comprehensive comparison between the 7-layer OSI Reference Model and the 4-layer TCP/IP Model. Explain how OSI layers 5, 6, and 7 map to TCP/IP.
4. **Switching Architectures:** Compare Circuit Switching, Message Switching, and Packet Switching across setup phases, delay, memory requirements, bandwidth efficiency, and real-world implementations.